

ECM Alternate Idle Signal Active when Not Requested

Symptoms

The ECM alternate idle signal is active when **not** requested.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.	
	STEP 1B. Check the alternate idle/auxiliary governor circuits for an open.	
	STEP 1C. Check the alternate idle/auxiliary governor circuits for a wire-to-wire short.	
	STEP 1D. Check the alternate idle/auxiliary governor circuits for a short to ground.	
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the alternate idle/auxiliary governor circuits for an open.	
	STEP 2B. Check the alternate idle/auxiliary governor circuits for a wire-to-wire short.	
	STEP 2C. Check the alternate idle/auxiliary governor circuits for a short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	DCU410 unit indicates fault(s)? YES	1B
	DCU410 unit indicates fault(s)? NO	1A-1

STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check the voltage at the battery 1 voltage (switched power) at the DCU410 unit. - Place one test on the battery 1 voltage (switched power) wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to OEM service manual.	Repair complete
	Less than +24-VDC? NO	1B

STEP 1B. Check the alternate idle/auxiliary governor circuits for an open.

Conditions : - Open the customer interface box. - Disconnect the alternate idle/auxiliary governor select signal wire at the X4 and C3 connectors.		
Action	Specification/Repair	Next Step
Check the alternate idle/auxiliary governor select signal wire for an open. - Place one test lead on the alternate idle/auxiliary governor select signal wire at the X4 connector. - Place the other test lead on the alternate idle/auxiliary governor select signal wire at C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1C
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1C. Check the alternate idle/auxiliary governor circuits for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the alternate idle/auxiliary governor select signal wire at the X4 connector.

Action	Specification/Repair	Next Step
Check the alternate idle/auxiliary governor select signal wire for a wire-to-wire short. • Place one test lead on the alternate idle/auxiliary governor select signal wire at the X4 connector. • Place the other test lead on all other wires at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the terminal strip. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	1D

STEP 1D. Check the alternate idle/auxiliary governor circuits for a short to ground.**Conditions :**

- Open the customer interface box.
- Disconnect the alternate idle/auxiliary governor select signal wire at the X4 connector.

Action	Specification/Repair	Next Step
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Check the alternate idle/auxiliary governor select signal wire for a short to ground. - Place one test lead on the alternate idle/auxiliary governor select signal wire at the X4 connector. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the terminal strip. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the alternate idle/auxiliary governor circuits for an open.

Conditions :

- Open the customer interface box.
- Disconnect the alternate idle/auxiliary governor select signal wire at the C3 connector and 50-pin ECM connector.

Action	Specification/Repair	Next Step

Check the alternate idle/auxiliary governor select signal wire for an open. - Place one test lead on the alternate idle/auxiliary governor select signal wire at the C3 connector. - Place the other test lead on the alternate idle/auxiliary governor select signal wire at 50-pin ECM connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 2B. Check the alternate idle/auxiliary governor circuits for a wire-to-wire short.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the alternate idle/auxiliary governor select signal wire for a wire-to-wire short. - Place one test lead on the alternate idle/auxiliary governor select signal wire at the C3 connector. - Place the other test lead on all other wires at the C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace connector. Refer to Procedure 019-208 (Deutsch®; HD20 and HD30 Connector Series) in Section 19 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System, Bulletin 4021493 or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the alternate idle/auxiliary governor circuits for a short to ground.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the alternate idle/auxiliary governor select signal wire for a short to ground. - Place one test lead on the alternate idle/auxiliary governor select signal wire at the C3 connector. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace connector. Refer to Procedure 019-208 (Deutsch®; HD20 and HD30 Connector Series) in Section 19 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System, Bulletin 4021493 or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location.